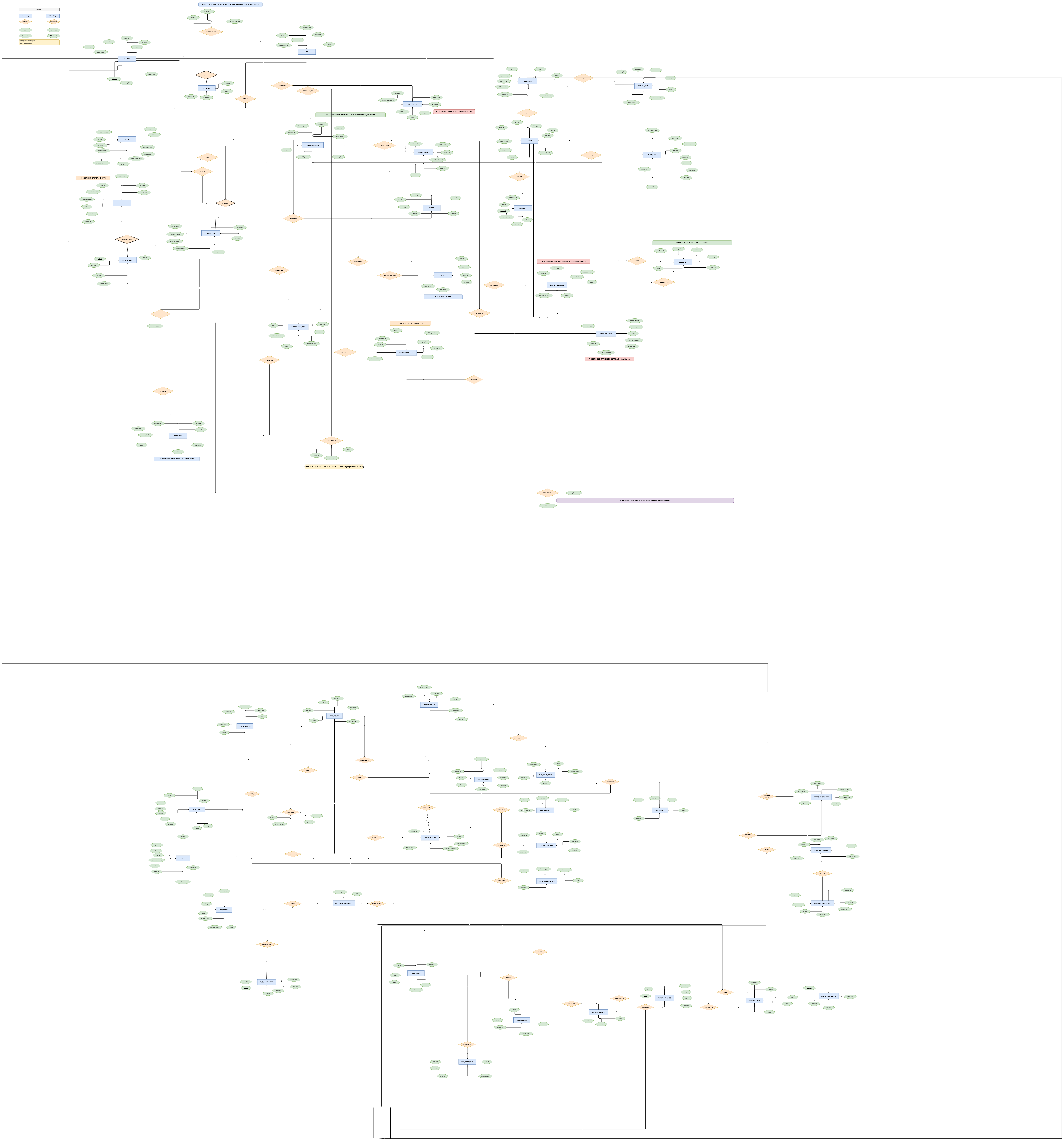
LEGEND
Strong Entity
Weak Entity
Relationship
Identifying Rel.
Attribute
Key Attribute
Derived Attr.
Multi-value Attr.
== Double line = Total Participation
— Single line = Partial Participation
1 / N / M = Cardinality labels
★ SECTION 1: INFRASTRUCTURE — Station, Platform, Line, Station-on-Line
sequence_no
is_active
dist_from_start_km
STATION_ON_LINE
zone_no
location
is_active
latitude
longitude
station_name
STATION
station_id
opening_date
station_type
HAS_PLATFORM
PLATFORM
direction
capacity
platform_no
is_available
total_length_km
line_id
color_code
line_name
operational_since
status
LINE
RUNS_ON
TRACKED_BY
SCHEDULED_ON
tracking_id
speed_kmph
dynamic_block_dist_m
LIVE_TRACKING
recorded_at
updated_ETA
longitude
latitude
★ SECTION 3: DELAY, ALERT & LIVE TRACKING
★ SECTION 2: OPERATIONS — Train, Train Schedule, Train Stop
departure_time
arrival_time
day_type
schedule_id
assigned_track_no
TRAIN_SCHEDULE
direction
schedule_status
normal_ETA
CAUSES_DELAY
delay_minutes
resolution_status
DELAY_EVENT
reported_at
affected_station_id
delay_id
reason
manufacturer
operational_status
train_id
TRAIN
train_type
train_number
commission_date
current_location
total_capacity
current_crowd_count
current_speed_kmph
is_on_time
RUNS
STOPS_AT
★ SECTION 4: DRIVER & SHIFTS
date_of_birth
driver_id
full_name
experience_years
joining_date
employment_status
DRIVER
salary
phone
license_no
ASSIGNED_SHIFT
HAS_STOP
stop_sequence
platform_no
scheduled_departure
TRAIN_STOP
is_active
scheduled_arrival
halt_duration_min
dynamic_ETA
message
alert_id
severity
alert_type
ALERT
is_resolved
created_at
GENERATES
shift_id
shift_end
DRIVER_SHIFT
shift_date
shift_start
working_hours
HAS_TRACK
direction
track_id
UNDERGOES
ASSIGNED_TO_TRACK
TRACK
length_km
is_active
track_number
track_status
★ SECTION 8: TRACK
HAS_CLOSURE
DRIVES
assignment_date
INVOLVED_IN
cost
MAINTENANCE_LOG
description
status
maintenance_date
log_id
maintenance_type
PERFORMS
★ SECTION 9: RESCHEDULE LOG
reason
original_dep_time
reschedule_id
new_dep_time
logged_at
old_track_no
HAS_RESCHEDULE
RESCHEDULE_LOG
new_track_no
done_by_emp_id
TRIGGERS
MANAGES
employee_id
full_name
joining_date
role
access_level
EMPLOYEE
email
department
salary
★ SECTION 7: EMPLOYEE & MAINTENANCE
TRAVELLING_IN
status
exited_at
boarded_at
★ SECTION 12: PASSENGER TRAVEL LOG — Travelling In (determines crowd)
FOR_JOURNEY
scan_timestamp
stop_role
★ SECTION 13: TICKET ↔ TRAIN_STOP (QR Entry/Exit validation)
full_name
email
passenger_id
phone
registered_at
PASSENGER
date_of_birth
disability_flag
passenger_type
HOLDS_PASS
pass_id
pass_type
valid_from
valid_to
TRAVEL_PASS
price
line_id_covered
institution_name
BOOKS
qr_code
ticket_type
ticket_id
issued_at
price_paid
from_station_id
TICKET
to_station_id
booking_channel
status
PRICED_BY
min_distance_km
fare_rule_id
max_distance_km
base_fare
FARE_RULE
normal_fare
senior_fare
effective_from
disabled_fare
child_fare
student_fare
PAID_VIA
payment_method
amount
PAYMENT
payment_id
transaction_ref
status
paid_at
★ SECTION 14: PASSENGER FEEDBACK
feedback_id
rating_1to5
comment
category
GIVES
FEEDBACK
status
submitted_at
FEEDBACK_FOR
★ SECTION 10: STATION CLOSURE (Temporary Removal)
closure_type
closure_id
start_datetime
end_datetime
STATION_CLOSURE
status
approved_by_emp
reason
incident_type
incident_datetime
location_desc
TRAIN_INCIDENT
status
incident_id
new_train_added_id
severity_level
reported_by_emp
★ SECTION 11: TRAIN INCIDENT (Crash / Breakdown)
normal_eta_mins
departure_time
arrival_time
day_type
BUS_SCHEDULE
schedule_status
schedule_id
route_number
route_id
route_name
route_type
BUS_ROUTE
is_active
total_length_km
operator_name
operator_id
operator_type
city
operator_code
BUS_OPERATOR
is_active
OPERATES
SCHEDULED_ON
RUNS
CAUSES_DELAY
min_distance_km
fare_rule_id
max_distance_km
child_fare
BUS_FARE_RULE
normal_fare
student_fare
senior_fare
effective_from
delay_minutes
reason
resolution_status
reported_at
BUS_DELAY_EVENT
delay_id
GENERATES
OWNED_BY
stop_code
stop_id
longitude
latitude
stop_name
BUS_STOP
stop_type
city
has_shelter
is_active
zone_no
ROUTE_STOP
sequence_no
is_active
dist_from_start_km
is_terminal
HAS_STOP
STOPS_AT
dynamic_eta
BUS_TRIP_STOP
is_active
scheduled_arrival
stop_sequence
scheduled_departure
incident_id
incident_type
severity_level
INVOLVED_IN
incident_datetime
BUS_INCIDENT
status
alert_id
alert_type
message
BUS_ALERT
severity
is_resolved
tracking_id
latitude
longitude
TRACKED_BY
BUS_LIVE_TRACKING
speed_kmph
updated_eta
recorded_at
log_id
maintenance_type
maintenance_date
UNDERGOES
BUS_MAINTENANCE_LOG
status
actual_cost
bus_type
bus_number
manufacturer
bus_id
BUS
current_crowd_count
current_lat
total_capacity
current_lng
operational_status
ASSIGNED_TO
license_no
full_name
driver_id
BUS_DRIVER
salary
experience_years
employment_status
phone
DRIVES
assignment_date
role
BUS_DRIVER_ASSIGNMENT
FOR_SCHEDULE
ASSIGNED_SHIFT
shift_date
BUS_DRIVER_SHIFT
working_hours
shift_id
shift_end
shift_start
shift_type
BOOKS
ticket_id
price_paid
status
BUS_TICKET
valid_to
qr_code
booking_channel
PAID_VIA
amount
paid_at
BUS_PAYMENT
status
payment_id
payment_method
SCANNED_AT
scan_role
BUS_STOP_SCAN
scan_id
is_valid
device_id
scan_timestamp
CONNECTS METRO
walking_dist_m
interchange_id
walking_time_min
INTERCHANGE_POINT
connection_type
is_covered
is_active
CONNECTS BUS
from_location
to_location
journey_id
COMBINED_JOURNEY
total_fare
PLANS
journey_type
total_eta_mins
HAS_LEG
from_stop_id
mode
leg_sequence
COMBINED_JOURNEY_LEG
to_stop_id
leg_fare
schedule_ref_id
leg_eta_mins
ON_SCHEDULE
TRAVELLING_IN
BUS_TRAVELLING_IN
status
exited_at
boarded_at
price
pass_type
valid_to
pass_id
BUS_TRAVEL_PASS
qr_code
HOLDS_PASS
valid_from
GIVES
feedback_id
category
rating
BUS_FEEDBACK
comment
FEEDBACK_FOR
status
config_key
BUS_SYSTEM_CONFIG
config_value
description
data_type